

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

***C02:** Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks:50 Annexure A

Analytical Problem Solving

Code of Conduct:

I VALLAPU MOHAN KUMAR_(192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

*I understand that any violation of academic integrity rules will result in disciplinary action. **Signature of the student:***

RUBRICS FOR CALCULATION

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	

Meth od Sele ction	20%	Selects most appropriate method/for mula with strong justification	Selects correct method with minor errors	Inappropri ate or partially correct method	Unable to select method	
Solut ion Proce ss	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorgan ized steps; multiple errors	No clear process	
Accu racy of Calcul ation	20%	All calculations accurate; correct final answer	Minor calculati on errors	Frequent errors; incorrect final answer	No valid calculati ons	
Interpr etati on of Resul ts	15%	Clearly interprets results with units and engineering relevance	Basic interpreta tion with minor omissions	Limited or unclear interpret ation	No interpret ation	

Annexure A

Analytical Problem Solving

Question 1: Salt-and-Pepper Noise Removal Before Edge Detection Scenario:

A grayscale image of size 1024×1024 is captured using a surveillance camera under poor lighting conditions. The image is corrupted by salt-and-pepper noise with a noise density of 0.05. This means that some pixels are randomly changed to either black or white intensity values. The image has to be preprocessed before applying an edge detection algorithm. Since edge detection is sensitive to abrupt intensity variations, the selected filter should remove noise effectively while preserving important edge details.

Answer the following:

- Identify the most suitable filtering technique for removing salt-and-pepper

noise without causing significant edge blurring.

Answer:

Median Filter (3×3) is the most suitable filter because it removes salt-and-pepper noise while preserving image edges.

- b. Explain why the selected filter performs better than an averaging filter for impulse noise removal.

Answer:

- Median filter replaces each pixel with the median value of its neighbors.
- It removes impulse noise without blurring edges.
- Averaging filter smooths the image by averaging pixel values, causing edge blurring.

- c. If a 3×3 median filter is applied, estimate the probability that a noisy center pixel will be corrected. Clearly state your assumptions.

Answer:

Probability that a noisy center pixel is corrected

Given:

- Noise density, $p = 0.05$
- Window size = 3×3 (9 pixels)

Assumption:

- Majority of neighboring pixels are noise-free
- Probability that all neighboring pixels are noise-free:

$$(1-0.05)^8=(0.95)^8\approx 0.663$$

Hence,

Probability of correcting the noisy center pixel $\approx 66.3\%$.

- d. Estimate the total number of noisy pixels present in the original image.

Answer:

Total number of noisy pixels

Image size: $1024 \times 1024 = 1,048,576$ pixels

Noise density = 0.05

$$1,048,576 \times 0.05 = 52,428.8$$

$\approx 52,429$ noisy pixels

- e. Estimate the expected number of noisy pixels remaining after median filtering.

Answer:

Assuming median filter removes 95% of noisy pixels,
Remaining noise

$$52,429 \times 0.05 = 2,621$$

Expected remaining noisy pixels $\approx 2,621$

- f. Explain how the preprocessing strategy would change if the same image was affected by Gaussian noise instead of salt-and-pepper noise.

Answer:

Use a Gaussian filter or Wiener filter instead of a median filter because Gaussian noise is distributed continuously rather than as impulse noise.

Question 2: Discretization of a Continuous Image

Scenario: An analog grayscale image has a smooth sinusoidal intensity variation along the horizontal direction. The intensity changes continuously with respect to the spatial coordinate x , while it remains constant along the y -direction. The image is defined over a spatial range of 0 to 255 in both directions. This continuous image has to be converted into a digital image for storage and further processing in a computer vision system.

Answer the following:

- a. Explain how the continuous image can be converted into a digital image using sampling and quantization.

Answer:

Conversion into a digital image

- **Sampling** converts continuous spatial coordinates into discrete pixels.
- **Quantization** converts continuous intensity values into discrete gray levels.

- b. Identify the spatial variation pattern of the image in the x -direction.

Answer:

Spatial variation pattern

The image has a **sinusoidal intensity variation** along the **x -direction** and remains constant along the **y -direction**.

- c. Determine the minimum sampling requirement needed to avoid aliasing in the x direction.

Answer:

According to the Nyquist Sampling Theorem,
Sampling frequency must satisfy

$$f_s \geq 2f_{max}$$

That is, sampling should be at least twice the highest spatial frequency

- d. Explain what will happen if the image is sampled below the required sampling rate.

Answer:

- Aliasing occurs.
 - Fine details become distorted.
 - False patterns appear in the image
- e. Suggest a suitable 8-bit quantization scheme for storing the image.

Answer:

Suitable 8-bit quantization

Use **8-bit grayscale quantization**.

Number of intensity levels:

$$2^8 = 256$$

Gray levels range from **0–255**.

- f. Discuss the effect of quantization on image quality, storage size, and intensity accuracy.

Answer:

Effect of quantization

- Reduces storage size.
- Slight loss of intensity accuracy.
- More quantization levels produce better image quality.

Question 3: Edge Detection in Thin Line Images

Scenario: A 512×512 grayscale image contains very thin vertical and horizontal white lines of 1-pixel thickness on a dark uniform background. Such images are commonly used in document analysis, engineering drawing interpretation, circuit diagram processing, and road-lane detection.

The objective is to detect the line boundaries accurately without losing thin edge information.

Answer the following:

- a. Compare Sobel, Prewitt, and Canny edge detectors for detecting 1-pixel-thick vertical and horizontal lines.

Answer:

Method	Advantages	Disadvantages
Sobel	Good edge detection	Slight edge thickening
Prewitt	Simple and fast	Less noise resistant
Canny	Best accuracy, thin edges	Higher computational cost

- b. For a horizontal edge located near pixel position (256, 256), compute the Sobel gradient response using the standard Sobel operator.

Answer:

For a horizontal edge,

Using Sobel kernels,

$$G_x = 0$$

$$G_y = \text{Maximum}$$

Gradient magnitude

$$G = \sqrt{G_x^2 + G_y^2} = |G_y|$$

Thus,

Horizontal edge gives a strong G_y response and nearly zero G_x

- c. Explain how the gradient magnitude and gradient direction help in identifying the strength and orientation of an edge.

Answer:

Gradient magnitude and direction

- Gradient magnitude indicates edge strength.
- Gradient direction indicates edge orientation

- d. Discuss the role of Gaussian smoothing in the Canny edge detection process.

Answer:

Role of Gaussian smoothing

Gaussian smoothing removes image noise before gradient calculation, reducing false edge detection.

e. Explain how Gaussian smoothing may affect edge localization, especially for thin lines.

Answer:

Effect on thin lines

- Slight edge blurring.
- Very thin edges may become weaker or disappear.

f. If the Canny threshold is reduced to half of its original value, explain the expected changes in detected edges, weak edges, and false edges.

Answer:

Reducing Canny threshold by half

- More edges detected.
- Weak edges become visible.
- False edges increase due to noise.

Question 4: Feature Extraction for Handwritten Digit Recognition Scenario:

A computer vision system is being developed for recognizing handwritten digits from scanned images. The handwritten digits vary in size, orientation, stroke thickness, and writing style. Therefore, robust feature extraction is required before classification.

Answer the following:

a. Explain the working principles of SIFT, SURF, and ORB feature extraction methods.

Answer:

Working principles

SIFT

- Detects stable keypoints using Difference of Gaussian.
- Generates scale and rotation invariant descriptors.

SURF

- Uses Hessian matrix approximation.
- Faster than SIFT.

ORB

- Combines FAST keypoint detector and BRIEF descriptor.
- Fast and suitable for real-time applications.

b. Compare SIFT, SURF, and ORB in terms of scale invariance, rotation invariance, computational cost, and real-time suitability.

Answer:

Feature	SIFT	SURF	ORB
Scale Invariant	Yes	Yes	Partial
Rotation Invariant	Yes	Yes	Yes
Speed	Slow	Medium	Fast
Real-Time	No	Moderate	Yes

- c. If a digit image is rotated by 45 degrees, identify which feature extraction methods can still produce reliable matching features.

Answer:

Rotated by 45°

Reliable matching methods:

- SIFT
- SURF
- ORB

- d. If the average number of keypoints detected per image is 120 and 50 images are processed, compute the total number of keypoints detected.

Answer:

Total keypoints

Given

120keypoints / image

50 images

$$120 \times 50 = 6000$$

Total keypoints = 6000

- e. Suggest a suitable dimensionality reduction technique to reduce the feature size while retaining 95% of the useful information.

Answer:

Dimensionality reduction

Principal Component Analysis (PCA)

It preserves approximately **95%** of useful information while reducing feature dimensions.

- f. Explain how dimensionality reduction improves classification efficiency in handwritten digit recognition.

Answer:

Benefits

- Faster classification.
- Lower memory usage.
- Reduced computation.
- Improved classifier performance.

Question 5: Morphological Preprocessing for Character Recognition

Scenario: A binary image of size 256×256 contains scanned handwritten characters. Due to poor scanning quality, the image contains small white noise dots in the background and small black gaps inside the character strokes.

Before applying character recognition, the image must be cleaned using morphological preprocessing techniques.

Answer the following:

- a. Identify the morphological operation suitable for removing small white noise dots from the background.

Answer:

Remove white noise dots

Opening operation

(Erosion followed by Dilation)

- b. Identify the morphological operation suitable for filling small gaps or holes inside the character strokes.

Answer:

Fill small gaps

Closing operation

(Dilation followed by Erosion)

- c. Explain the difference between opening and closing operations using simple image processing terminology.

Answer:

Difference

Opening

- Removes small white objects.
- Preserves larger objects.

Closing

- Fills small holes.

- Connects broken regions.
- d. If a 3×3 square structuring element is used, explain how it affects thin character strokes.

Answer:

Effect of 3×3 structuring element

- Removes tiny noise.
 - Preserves most character strokes.
 - Very thin strokes may become slightly thinner.
- e. Discuss the risk of using a large structuring element during preprocessing.

Answer:

Risk of large structuring element

- Loss of fine character details.
 - Characters may merge.
 - Recognition accuracy decreases.
- f. Propose a complete preprocessing pipeline for handwritten character recognition, starting from image acquisition to feature extraction.

Answer:

- Image acquisition.
- Grayscale conversion.
- Binary thresholding.
- Noise removal using Opening.
- Gap filling using Closing.
- Edge or feature extraction (SIFT/ORB).
- Dimensionality reduction (PCA).
- Character classification.